

EMMANUEL OPPONG

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Education

Grambling State University

Bachelor of Science in Computer Science | GPA: 3.93

Graduating, May 2028

Grambling, Louisiana

Relevant Coursework: Object-Oriented Programming, Software Engineering, Data Structures & Algorithms, Operating Systems

Awards: Dean's List 24 and 25, **Presidential Scholarship**

Experience

Software Engineer Apprentice

May 2025 – August 2025

Ericsson

Remote

- Engineered a Kafka-streaming microservice (**Kubernetes, REST API**) processing **400K+** daily events across **6** topics, replacing manual SQL workflows and achieving sub-**200ms** latency.
- Architected an event-driven system (**Apache Kafka, F#, S3**) aggregating **10K+** pricing records into **RESTful** endpoints, enabling real-time access to previously disconnected historical trade data.
- Built a **React** dashboard visualizing distributed pipeline timelines in real time, reducing debugging and resolution time from **60 minutes** to **10 minutes** for a 30-engineer team.

Software Engineer Intern

June 2023 – September 2023

Noon

Abu Dhabi, UAE

- Developed a scalable IoT data pipeline (**Kafka Connect, MQTT, PostgreSQL**) streamlining sensor data processing and reducing downtime detection by **40%**.
- Created **50+** automated unit & integration tests (**SpringBoot, JUnit, Postman**), improving backend reliability by **80%** through rigorous edge-case handling in concurrent environments.
- Implemented a real-time push notification system (**Google Firebase**) delivering status updates with under **10ms** latency for optimized device monitoring.

Projects

MedStroke AI | *Python, TypeScript, FastAPI, TensorFlow, React*

- Developed a full-stack stroke diagnosis system with real-time monitoring, role-based access, NIHSS scoring, and automated tPA eligibility to cut diagnostic delays.
- Integrated a **Random Forest Machine Learning model** to analyze blood sample data, achieving **45%** accuracy in stroke risk prediction and providing neurologists with automated, data-driven diagnoses.
- Piloted system at a hospital in Louisiana, improving early stroke detection and reducing time-to-treatment by **35%**, directly enhancing patient outcomes and care coordination.

Go Container | *Go, Linux*

- Built a **container runtime in Go** with process isolation by leveraging Linux namespaces to give each container its own process space, filesystem, and hostname without affecting the host.
- Implemented **resource controls using cgroups** to cap the number of processes and automatically clean up after container exit, ensuring reliable and efficient system utilization.
- Set up a secure, self-contained filesystem environment using **chroot and mounting pseudo-filesystems**, providing complete separation from the host and preventing unwanted interference.

NanoSQL | *C*

- Built a **high-performance SQLite-clone database engine in C**, implementing **B-tree indexing**, persistent storage, and custom memory management.
- Engineered a **command-line interface with efficient data serialization**, parameterized SQL operations (INSERT/SELECT), and integrity validation.
- Architected a **complete B-tree implementation featuring dynamic node splitting**, rebalancing, and optimized page allocation.

Technical Skills

Languages: Java, C++, C, TypeScript, Python, Dart, JavaScript, SQL, Protocol Buffers

Technologies/Frameworks: ReactJS, VueJS, SpringBoot, Flutter, Flask, MongoDB, NodeJS, Pandas, NumPy, CI/CD, TensorFlow

Tools: GitHub, Figma, Kubernetes, Linux, Grafana, Kafka, Bash

Concepts: Parallel Computing, Full-Stack Development, Mobile Development, Microservices, Agile Methodologies

Activities / Leadership

National Society of Black Engineers

January 2025 – Present

Assistant Vice-President

Grambling State University

- Reignited the love for engineering to support Computer Science majors, building a community of 300+ members and securing partnerships from technical companies worth **\$3K**

ColorStack

September 2024 – Present

Member

Grambling State University

- Winner of the 2024 Winter Hackathon, out of 1000 participants worldwide.